

Financial Time Series Analysis

Lecture 14

- ▶ **The practice ProctorTrack quiz is ready to be taken. 23 students are not yet authorized for ProctorTrack and MUST take it in order to take the final exam on Monday**
- ▶ TA Session: Friday, May 7, 11:00am-12:00pm
- ▶ Final Exam: Monday, May 10, 10:00am
- ▶ Please take the course evaluation (for all classes).
Deadline is Thursday 11:59pm.

Outline

- ▶ State Space Models
 - ▶ Bayesian approach
 - ▶ “Filtering in Finance” Heston stochastic volatility model
 - ▶ Stochastic volatility model

State Space Models: 2

- ▶ The two noise processes, $\mathbf{V}_t$ and $\mathbf{W}_t$ are assumed to be $\text{WN}(\mathbf{0}, \boldsymbol{\Sigma}_{X,t})$ and $\text{WN}(\mathbf{0}, \boldsymbol{\Sigma}_{Y,t})$ respectively.
- ▶ The two noise processes are uncorrelated, i.e. $E(\mathbf{V}_s \mathbf{W}_t) = \mathbf{0}, \forall s, t.$
- ▶ The standard time series models (AR, MA, ARMA) have state space representations.

State Space Models: 5

► Prediction or Forecasting

Estimate $\mathbf{X}_{t+1}$ or $\mathbf{X}_{t+k}$ using $\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t$. This can be denoted $\hat{\mathbf{X}}_{(t+k)|t}$

► Filtering

Estimate $\mathbf{X}_t$ using $\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t$. We denote this $\hat{\mathbf{X}}_{t|t}$

► Smoothing

Estimate $\mathbf{X}_t$ given $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_T$, for $1 \leq t < T$. This is denoted $\hat{\mathbf{Y}}_{t|T}$ indicating the estimate of $\mathbf{X}_t$ conditional on $\mathcal{F}_T$.

Bayesian Background to Kalman Methods: 1

- ▶ Recall the Bayesian approach to statistics: unknown parameters have probability distributions (prior distribution).
- ▶ When one has data that was generated by a mechanism involving the unknown parameters, this generates a likelihood function, $f(\text{data}|\text{parameters})$.
- ▶ The (prior) distribution of the parameters coupled with the likelihood function can be combined to generate a new (posterior) distribution of the parameters using Bayes theorem.

$$P(\text{parameters}|\text{data}) = \frac{P(\text{parameters}) * P(\text{data}|\text{parameters})}{P(\text{data})}.$$

Bayesian Background to Kalman Methods: 2

- ▶ If minimum mean square error is the criterion, then given only the prior distribution, the minimum mean square error predictor of the parameters is the mean of the prior distribution. The associated minimum mean square error is the variance of the prior distribution, $\text{Var}(\text{parameters})$.
- ▶ Given the data, the minimum mean square error predictor is the mean of the posterior distribution, and the associated mean square error is the variance of the posterior distribution, $\text{Var}(\text{parameters}|\text{data})$.
- ▶ These ideas are best understood with a simple example using the normal distribution.

Bayesian Background to Kalman Methods: 3

- ▶ Suppose that there is an unknown parameter, θ . We have a prior distribution for θ given by $N(\mu, V_0^2)$. We generate data, $X_1, X_2, \dots$ that are i.i.d. $N(\theta, \sigma^2)$. Suppose the data arrives one at a time.
- ▶ After observing X_1 we can update the prior to a posterior,

$$\theta|X_1 = x_1 \text{ has a } N(\mu_1, V_1^2),$$

where

$$\mu_1 = \frac{\sigma^2}{\sigma^2 + V_0^2} \mu + \frac{V_0^2}{\sigma^2 + V_0^2} X_1$$

$$V_1^2 = \frac{\sigma^2}{\sigma^2 + V_0^2} V_0^2.$$

Bayesian Background to Kalman Methods: 4

This easily extend to $X_1, \dots, X_n$ i.i.d. $N(\theta, \sigma^2)$. We can reduce to the sufficient statistic $\bar{X}_n$, which has $N(\theta, \sigma^2/n)$. Thus the posterior distribution is $N(\mu_n, V_n^2)$ and

$$\mu_n = \frac{\sigma^2}{\sigma^2 + nV_0^2} \mu + \frac{nV_0^2}{\sigma^2 + nV_0^2} \bar{X}_n,$$

$$V_n^2 = \frac{\sigma^2}{\sigma^2 + nV_0^2} V_0^2.$$

Equivalently, the updating can be done sequentially (on-line)

$$\mu_n = \frac{\sigma^2}{\sigma^2 + V_0^2} \mu_{n-1} + \frac{V_0^2}{\sigma^2 + V_0^2} X_{n-1}$$

$$V_n^2 = \frac{\sigma^2}{\sigma^2 + V_{n-1}^2} V_{n-1}^2.$$

Bayesian Background to Kalman Methods: 5

What is the relevance to the state space models and the Kalman methodology? We simplify by considering the univariate case, although everything carries over to the multivariate case.

- ▶ In the state space model we have a dynamic version of the problem. That is, there is an unknown state X_t that is changing with time according to the process

$$X_{t+1} = FX_t + V_t,$$

where V_t has $N(0, Q_t)$ distribution.

- ▶ At each step, there is an observation, Y_t given by

$$Y_t = GX_t + W_t,$$

where W_t has a $N(0, R_t)$ distribution.

Bayesian Background to Kalman Methods: 6

- ▶ We begin by assuming that there is an initial normal prior distribution for X_1 .
- ▶ At stage $t-1$, we have observed $Y_1, \dots, Y_{t-1}$ and using those data, we have a posterior distribution for X_{t-1} .
- ▶ Unlike the previous example, this must be done a step at a time because the X_t process is dynamic, not just an unknown constant.
- ▶ Nevertheless, each step can be carried out using the Bayesian machinery.
- ▶ Recall Glasserman pages on joint, marginal and conditional distributions of multivariate normals.

Bayesian Background to Kalman Methods: 7

- ▶ To do Kalman filtering, we want to estimate X_t conditional on $\mathcal{F}_t = \sigma(Y_1, \dots, Y_t)$.
- ▶ We have the posterior distribution for X_{t-1} given $\mathcal{F}_{t-1}$, let's say it is $N(\mu_{t-1}, \sigma_{t-1}^2)$.
- ▶ Using the evolution of the state equation, the conditional distribution of X_t given X_{t-1} is given by $N(FX_{t-1}, Q_t)$
- ▶ Also the conditional distribution of $Y_t|X_t$ is given by the observation equation and is $N(GX_t, R_t)$.
- ▶ Now we can consider the triplet (X_{t-1}, X_t, Y_t) given $\mathcal{F}_{t-1}$.

Bayesian Background to Kalman Methods: 8

- ▶ This triplet has a trivariate normal distribution, the marginal of X_{t-1} , which is the posterior given $\mathcal{F}_{t-1}$ times the conditional of $X_t|X_{t-1}$ times the conditional of $Y_t|X_t$.
- ▶ For Kalman filtering, we want the conditional distribution of X_t given Y_t (note: $\mathcal{F}_{t-1}$ is incorporated in the posterior of X_{t-1}).
- ▶ We can reduce this to the bivariate distribution of (X_t, Y_t) , which we have, and we seek to generate the conditional distribution of $X_t|Y_t$.
- ▶ This requires Bayes formula or the Glasserman formulas.

Bayesian Background to Kalman Methods: 9

- ▶ Recall if X_{t-1} has a $N(\mu_{t-1}, \sigma_{t-1}^2)$ distribution and $X_t|X_{t-1} = FX_t + V_t$ which has a $N(FX_t, Q_t)$ distribution, then
- ▶ the distribution of marginal distribution of X_t is $N(F\mu_{t-1}, F^2\sigma_{t-1}^2 + Q_t)$
- ▶ So we begin with this as the prior distribution for X_t and seek to find the posterior of X_t given Y_t , the Kalman filter.

The Kalman Recursions

- ▶ In all three cases there are **Kalman recursions** to solve these problems based on minimizing mean square error. The recursions allow one to solve these problems “on-line,” that is in the case of forecasting and filtering to use the current results and the new data to obtain new estimates/forecasts without needing the entire data set that preceeded it.

Kalman Prediction: 1

- ▶ In Kalman prediction, the goal is to predict/forecast $\mathbf{X}_{t+1}$ given $(\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t)$, a future value of $\mathbf{X}$ given the information from $\mathbf{Y}$ up to time t .
- ▶ We let the mean squared error of the prediction be denoted by

$$\Omega_{t+k} = E((\mathbf{X}_{t+k} - \hat{\mathbf{X}}_{t+k})(\mathbf{X}_{t+k} - \hat{\mathbf{X}}_{t+k})^T).$$

- ▶ The Kalman prediction recursion begins with $\hat{\mathbf{X}}_1$ and proceeds one-step at a time to $\hat{\mathbf{X}}_{t+1}$, as follows:

Kalman Prediction: 2

$$\hat{\mathbf{X}}_1 = E(\mathbf{X}_1 | \mathbf{Y}_0),$$

$$\hat{\mathbf{X}}_{t+1} = \mathbf{F}_t \hat{\mathbf{X}}_t + \mathbf{\Theta} \mathbf{\Delta}_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t),$$

$$\mathbf{\Omega}_{t+1} = \mathbf{F}_t \mathbf{\Omega}_t \mathbf{F}_t^T + \mathbf{Q}_t - \mathbf{\Theta}_t \mathbf{\Delta}_t^{-1} \mathbf{\Theta}_t^T,$$

where

$$\mathbf{\Delta}_t = \mathbf{G}_t \mathbf{\Omega}_t \mathbf{G}_t^T + \mathbf{R}_t,$$

$$\mathbf{\Theta}_t = \mathbf{F}_t \mathbf{\Omega}_t \mathbf{G}_t^T.$$

Kalman Filtering

- ▶ For Kalman filtering, we want to predict/estimate the current state, $\mathbf{X}_t$ given $(\mathbf{Y}_0, \dots, \mathbf{Y}_t)$. This is denoted by $\mathbf{X}_{t|t}$.
- ▶ The estimate is a combination of the one-step-ahead prediction from time $t - 1$, $\hat{\mathbf{X}}_t$ and the difference between the observed $\mathbf{Y}_t$ and its predicted value using the observation equation, $\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t$

$$\mathbf{X}_{t|t} = \hat{\mathbf{X}}_t + \mathbf{\Omega}_t \mathbf{G}_t^T \mathbf{\Delta}_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t),$$

where the relevant terms were defined earlier.

Stochastic Volatility: 1

- ▶ Recall we have develop ARCH/GARCH modeling to create models that better conform to the stylized facts of financial markets: heteroscedasticity, volatility clustering, heavy tail behavior.
- ▶ The GARCH model created this behavior “within itself,” not based on external events/shocks. Yet we have seen that many stocks (particularly those impacted by the same factors, such as the market, industry sectors, etc) exhibit similar behavior with respect to volatility clustering.
- ▶ An alternative model would allow for exogenous processes to impact the volatility behavior of a financial asset.
- ▶ Such models are called stochastic volatility models, and a state space formulation of these models allows us to employ “Kalman methodology” to filter and predict.

Stochastic Volatility: 2

- ▶ In state space models, there is an unobserved or latent variable(s) modeled and an observation/measurement process. We often want to make inference about the latent process.
- ▶ A very good example of this situation is volatility. Volatility is unobserved, yet crucial to the valuation of assets (e.g. an option price).
- ▶ We want to model volatility as the unobserved state and then have an observation process that can be used to estimate (filter) and predict the latent process.
- ▶ An example using the Heston model is presented in the paper “Filtering in Finance.”

Stochastic Volatility: 3

- ▶ The Heston model involves the state process, v_t , and the observation process, S_t , defined by the SDE:

$$dv(t) = (\omega - \theta v_t)dt + \xi \sqrt{v_t} dW_t^v,$$

$$dS_t = rS_t dt + \sqrt{v(t)} S_t dW_t^S.$$

- ▶ The volatility process is a CIR process, whereas the stock price process is GBM with stochastic volatility where:
 - ▶ r is the risk free rate,
 - ▶ θ is the speed of mean reversion,
 - ▶ ω/θ is the long-run equilibrium variance
 - ▶ ξ is the volatility of the variance process.
 - ▶ ρ is the correlation between W_t^v and W_t^S
- ▶ Conditional on v_t , the price process is a GBM

Stochastic Volatility: 4

- ▶ The Heston model is defined in continuous time, and we need to convert it to discrete time, and put it into state space form.
- ▶ The parameter set $(\omega, \theta, \xi, \rho)$ must be calibrated with market data to have the model in risk-neutral form.
- ▶ We can rewrite the stock price equation

$$\log S_{t+\Delta} = \log S_t + (r - .5v_t)\Delta + \sqrt{v_t}\Delta Z_t^S.$$

- ▶ We next use an Euler discretization for the two SDEs with time step Δ :

Stochastic Volatility: 5

$$\log S_{(i+1)\Delta} = \log S_{i\Delta} + (r_{i\Delta} - .5v_{i\Delta})\Delta + \sqrt{v_{i\Delta}\Delta}Z_{(i+1)\Delta}^S,$$

$$v_{(i+1)\Delta} = v_{i\Delta} + (\omega - \theta v_{i\Delta})\Delta + \xi \sqrt{v_{i\Delta}\Delta}Z_{(i+1)\Delta}^v,$$

where $\{(Z_i^S, Z_i^v), 1 \leq i \leq N\}$, $\Delta = T/N$, are iid standard bivariate normal random vectors with correlation ρ .

- It is possible to eliminate the correlation parameter ρ by introducing the independent pair $(\tilde{Z}_i^v, Z_i^S)$ where

$$Z_i^v = \rho Z_i^S + \sqrt{1 - \rho^2} \tilde{Z}_i^v$$

and substituting $\tilde{Z}_{i\Delta}^v = \frac{1}{\sqrt{1-\rho^2}}(Z_{i\Delta}^v - \rho \tilde{Z}_{i\Delta}^S)$ into the volatility equation.

Stochastic Volatility: 6

- ▶ Substituting $\sqrt{v_k}\Delta Z_i^S = \log(S_{(i+1)\Delta}/S_{i\Delta}) - (r_{i\Delta} - .5v_{i\Delta})\Delta$ into the volatility equation, we arrive at the state space version of the Euler discretized Heston model with independent noise terms:

$$\begin{aligned}v_{(i+1)\Delta} &= v_{i\Delta} + [\omega - \rho\xi r_{i\Delta} - (\theta - .5\rho\xi)v_{i\Delta}]\Delta \\ &+ \rho\xi \log(S_{(i+1)\Delta}/S_{i\Delta}) + \xi\sqrt{1 - \rho^2}\sqrt{v_{i\Delta}}\Delta\tilde{Z}_k^v.\end{aligned}$$

for the state equation and

$$\log S_{(i+1)\Delta} = \log S_{i\Delta} + (r_{i\Delta} - .5v_{i\Delta})\Delta + \sqrt{v_{i\Delta}}\Delta Z_k^S,$$

for the measurement equation.

Stochastic Volatility: 7

- ▶ Once the stochastic volatility model is in state space form, we can utilize the Kalman procedures for statistical inference.
- ▶ In the particular case, the volatility is of great interest, and inferring the current volatility using Kalman filtering has significant implications (e.g. in valuation of options, comparing current volatility estimates with implied volatility)
- ▶ There are a number of stochastic volatility models, e.g. the Hull-White model we used in the simulation course.
- ▶ Estimation of these models is known to be more complicated than GARCH models.